



Infrared dim-small target tracking via singular value decomposition and improved Kernelized correlation filter



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HIGHLIGHTS

- The singular value decomposition based algorithm can enhance a target effectively.
- The curvature filter preserves the edge and eliminates the noise in base samples.
- Target tracking is translated into a detection task, reaching a high speed.
- This algorithm yields an outstanding performance over kinds of complex backgrounds.

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ABSTRACT

Infrared small target tracking plays an important role in applications including military reconnaissance, early warning and terminal guidance. In this paper, an effective algorithm based on the Singular Value Decomposition (SVD) and the improved Kernelized Correlation Filter (KCF) is presented for infrared small target tracking. Firstly, the super performance of the SVD-based algorithm is that it takes advantage of the target's global information and obtains a background estimation of an infrared image. A dim target is enhanced by subtracting the corresponding estimated background with update from the original image. Secondly, the KCF algorithm is combined with Gaussian Curvature Filter (GCF) to eliminate the excursion problem. The GCF technology is adopted to preserve the edge and eliminate the noise of the base sample in the KCF algorithm, helping to calculate the classifier parameter for a small target. At last, the target position is estimated with a response map, which is obtained via the kernelized classifier. Experimental results demonstrate that the presented algorithm performs favorably in terms of efficiency and accuracy, compared with several state-of-the-art algorithms.

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1. Introduction

The infrared target tracking has been widely used in military and civilian. Similar to visual tracking, the key step of infrared target tracking is to constantly estimate the target position in infrared sequences. When a target is far away from the infrared system, the dim target with only a few pixels easily submerges in the cluttered background with strong radiation. Generally, the performance of real-time and accuracy is the crucial challenge for infrared dim target tracking as the target moves fast [1–5]. Besides, the edge of cloud around a target can lead to tracking migration. Therefore, the accurate detection and tracking of a dim target are considered as a difficult work.

Several researchers have made contributions to the infrared dim target detection and tracking [3–9], such as Template Matching Tracking (TMT) algorithm [3,4], Mean Shift (MS) algorithm [5,6], Temporal Spatial Fusion Filtering (TSFF) algorithm [7], and Particle Filter (PF) algorithm [8,9]. Specifically, the TMT algorithm completed matching between frames using the gray feature. The TSFF algorithm processed the background suppression with the top-hat filter in the spatial domain. Meanwhile, it took advantage of an improved frame difference method to enhance the target. Besides, the MS algorithm represented infrared target appearances by kernel weighted gray histogram and utilized the mean shift procedure to identify the most likely position of the target in the next frame. To improve the performance of these tracking algorithms, Liu et al. presented a tracking framework based on the template matching combined with the Kalman Filter (KF) [10]. Such framework used the projection coefficients of PCA as templates and measured the matching degree by using nonlinear correlation

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coefficients. However, the tracking migration is frequent as the background clutter is sever. Li et al. combined an adaptive KF with mean shift [11]. The center of the object predicted by KF was utilized as the initial value of the MS algorithm. The searching result of MS was fed back as the measurement of the adaptive KF. Besides, the estimate parameters of KF were adjusted by the Bhat-tacharyya coefficient adaptively. However, it is possible for these algorithms to lose the dim moving target with a low Signal-to-Noise Ratio (SNR). Moreover, in PF-based algorithms, target tracking is considered as a recursive Bayesian estimate problem, and the estimated state is represented by some parameters of a tracked target, such as positions and velocity [12,13]. Huang et al. proposed an efficient algorithm of the small object position [12], it's initialized using a strong detector which is created from the shape analysis of foreground blobs and a particle filter-based tracker can handle the ambiguity of the template matching. Easily, particles can degrade into a single value. Li et al. presented a PF-based tracking algorithm to cope with the issue of the uncertainty in the dim target tracking [13]. The discriminative over-complete dictionary is utilized to enlarge the difference between the target particle and the background particle. Further, the particle discriminative sparse representation can improve the accuracy of target motion estimation by means of heightening the weight of target particles. Nevertheless, this algorithm don't focus on obtaining a sufficient account of a target's feature.

Recently, a tracking mechanism containing an online training classifier was presented [14–19]. Zhang et al. [14] proposed a Fast Compressive Tracking (FCT) algorithm, which constructed a very sparse measurement matrix. Then, the tracking task was formulated as a binary classification via a naive Bayes classifier. Nevertheless, the classifier got a mistake as the small target submerged in the background clutter. To track an infrared target, Li et al. combined the appearance model in the FCT algorithm with the popular ℓ_1 tracker [15]. Moreover, Zhang et al. packed the correlation filter as the probability theory, and utilized the dense sampling to track an area target [16]. The Spatio-Temporal Context (STC) tracker obtained a target confidence map, which meant tracking was converted to a detection problem. Then, He et al. presented a tracking algorithm based on the low-rank representation and a weighted correlation filter [17], which integrated a multi-features weight function into the correlation filter in the tracking procedure. Yet, the detection procedure took more time to obtain a detection result as the image size became larger. In order to track the infrared dim moving target, Qian et al. combined the STC tracker with the Guided Image Filter (GIF) algorithm [18], helping to eliminate the negative effect caused by the background clutter. Nevertheless, this algorithm is sensitive to the target speed. Besides, the Kernelized Correlation Filter (KCF) tracker [19] utilized the dense sampling to make up for the deficiency of the target intensity. This tracker trained the classic Support Vector Machine (SVM) classifier in the Fourier domain, which was the reason that it ran at a high speed. However, for the infrared small target tracking, the KCF algorithm drifts to the background easily, due to the fact that the background contains much edge.

Moreover, to efficiently extract the discriminative feature for the tracking mechanism, some researchers present preprocessing algorithms, such as image filtering-based algorithm [20,21], saliency detection algorithm [22,23], and background subtraction algorithm [24,25]. Bae proposed a spatial and temporal Bilateral Filter (BF) algorithm [20], which predicts the spatial infrared background using a spatial BF, and a temporal background profile using a temporal BF. Therefore, it extracts a trajectory for small targets based on the differences between an original infrared image. However, it is time-consuming for target tracking. Wan et al. [23] proposed a sparse representation model with a non-negative

constraint to detect the dim target. For less computation, a preprocessing method based on frequency saliency detection is adopted to extract the suspected regions containing the target. Nevertheless, kinds of complex backgrounds result in an inaccurate parameter estimation in the tracking model. Liu et al. built a background image with the pixel's neighborhood mean value [24], and then calculated the difference between the current frame and its corresponding background frame with an update scheme. Easily, the mean filter is influenced by the background clutter. Yet, this local filter can be replaced by the Singular Value Decomposition (SVD) [26], which is an efficient global method to reconstruct the background.

Therefore, an algorithm based on an the Singular Value Decomposition (SVD) [26–27] and an improved KCF [19] is presented for infrared small target tracking. Firstly, the complex background is estimated via the efficient SVD-based algorithm, and the target is enhanced by subtracting the corresponding estimated background with update from the original image. Moreover, the enhanced image is utilized as an input of the KCF-based algorithm. In the KCF algorithm, the dense sampling and the online classifier are capable of extracting sufficient feature information. Besides, the Gaussian Curvature Filter (GCF) [28] algorithm weakens the negative influence of the edge in the base sample, helping to obtain the classifier parameter for a small target. Finally, target position is estimated by finding the maximum value in a response map. Experimental results show that the presented algorithm performs favorably in terms of accuracy and speed.

The remainder is organized as follows. In Section 2, the procedure of the target enhancement part is given. In Section 3, the framework of the KCF-based algorithm is stated first. Meanwhile, the excursion problem is analyzed. Moreover, a high-speed GCF algorithm is adopted to improve the KCF tracker, and then the SVD-IKCF is presented. In Section 4 the performance of the presented algorithm is tested over several infrared sequences, compared with several state of the art algorithms. Section 5 provides the conclusions of this work.

2. SVD-based target enhancement

Generally, it is difficult to extract the feature information for an infrared small target with a low SNR. Now, an efficient algorithm based on the SVD [26,27,29] is designed for enhancing small targets. Given an $M \times N$ image matrix R_t with rank r , it can be factorized into three matrices

$$R_t = U \begin{pmatrix} \Sigma_r & 0 \\ 0 & 0 \end{pmatrix} V^T \quad (1)$$

where U and V^T (the conjugate transpose of V) are $M \times M$ and $N \times N$ unitary matrices, which are called the left-singular and the right-singular vectors of R_t , respectively. $\Sigma_r = \text{diag}(\lambda_1, \lambda_2, \dots, \lambda_r)$ is a diagonal matrix with non-negative real numbers on the diagonal, and $\lambda_i (i = 1, \dots, r)$ are known as singular values that are in descending order, i.e., $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_r \geq 0$.

Intuitively, singular values represent the image information. A large value means that its corresponding component is more inclined to reflect the image contour information. Similarly, a small singular value refers to the image detail. This property helps to estimate the background by eliminating several small singular values. At frame t , about c percent of the sum for all singular values is set as a threshold to determine the singular value's number. Therefore, the estimated background B_t is reconstructed with the k selected values,

$$B_t = U \begin{pmatrix} \Sigma_k & 0 \\ 0 & 0 \end{pmatrix} V^T \quad (2)$$

where $\sum_k = \text{diag}(\lambda_1, \lambda_2, \dots, \lambda_k)$ is a diagonal matrix with non-negative real numbers on the diagonal and k satisfy $\sum_{i=1}^k \lambda_i / \sum_{i=1}^r \lambda_i \approx c/100$.

Then, the difference image D_t is obtained by subtracting the corresponding background image B_t from the current frame R_t ,

$$D_t = |R_t - B_t| \quad (3)$$

In order to constantly adapt to the environment, the background image is updated through an efficient update mechanism [24], which utilizes the former and the current frame with the weight factor γ ,

$$B_{t+1} = (1 - \gamma) \cdot B_t + \gamma \cdot B_{t+1} \quad (4)$$

Further, a group of experimental results is shown in Figs. 1–3.

As shown in Fig. 1(c) and (e), the difference between the original infrared image and the background varies largely. It is shown in Fig. 1(d) that the target region is salient, for the reason that the background is updated in every frame. In Fig. 2, since the image background changes largely, the target in the difference image (see Fig. 2(d)) is surrounded with much noise. Nevertheless, the SNR of the dim target (marked by the red ellipse) increases. Moreover, in Fig. 3(a), the target with a low SNR submerges in the cluttered background. The target's intensity keep to be weak even if it has been enhanced, which is shown in Fig. 3(d) and Fig. 3(e). However, it is better to adopt the enhanced image sequence instead of the original infrared sequence. In summary, the SVD-based algorithm is appropriate for estimating the infrared background and small targets can be well enhanced via an update mechanism. Besides, the SVD is superior to several other background suppression algorithms in the speed. Further, the enhanced infrared image sequence is utilized as an input for the target tracking in the following section.

3. The improved KCF algorithm

3.1. The KCF algorithm

The key of the KCF algorithm is to train a classifier though a ridge regression, which is to find a function $f(z) = w^T z$ that minimizes the squared error over samples x_i and their regression targets y_i . The objective function of the linear regression is formulated as

$$\min_w \sum_i (f(x_i) - y_i)^2 + \lambda \|w\|^2 \quad (5)$$

where λ is a regularization parameter, and w denotes the regression coefficient. $\|\cdot\|$ and $\sum$ denote the norm operation and the sum operation, respectively.

The kernel trick [30] is utilized to improve the performance further, by allowing the classification on a rich high-dimensional feature space. The inputs are mapped to the feature space using $\phi(x_i)$, defined by the kernel $k(x_j, x_i) = \phi(x_j) \cdot \phi(x_i) = x_j \cdot x_i$. Generally, w can be

expanded as a linear combination of the inputs, i.e., $w = \sum_i \alpha_i \phi(x_i)$, where α_i is a linear coefficients.

Then, the solution to the kernelized version of the ridge regression [19] is expressed as

$$\alpha = (K + \lambda I)^{-1} y \quad (6)$$

where I is the identity matrix, and K denotes the kernel matrix with elements $K_{ij} = k(x_i, x_j)$. The solution w is implicitly represented by the coefficients α (has the element α_i). In the current frame, y (has the element y_i) represents a prior of the target position, and can be modeled as [18]

$$y = b \cdot \exp(-|D/\sigma_1|^\beta) \quad (7)$$

where $\exp(\cdot)$ denotes the exponential function, and b is a normalization constant. D denotes the Euclidean distance between the target and one pixel in the neighborhood. σ_1 and β represent a scale parameter and a shape parameter, respectively.

3.1.1. Circulant matrix

The dense sampling technology is utilized to obtain a circulant sample matrix X , which exploits the target structure efficiently, and is defined as

$$X = C(x_1) = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} v_1 & v_2 & v_3 & \cdots & v_n \\ v_n & v_1 & v_2 & \cdots & v_{n-1} \\ v_{n-1} & v_n & v_1 & \cdots & v_{n-2} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ v_2 & v_3 & v_4 & \cdots & v_1 \end{bmatrix} \quad (8)$$

where $x_1 = [v_1, v_2, v_3, \dots, v_n]$ is the gray intensity vector of the base sample, which denotes a positive sample. Besides, x_i ($i = 2, 3, \dots, n$) serves as the gray intensity vector of the negative sample, which is obtained by shifting x_1 with $(i - 1)$ elements. n represents the pixel number of one sample. $C(x_1)$ means that the circulant matrix X is generated with the base sample x_1 . For convenience, the size of a sample is set as the integral multiple of a target template, considering the background information around the target.

Obviously, the sample matrix X is an $n \times n$ matrix, which contains abundant information. Several samples in X are shown in Fig. 4.

As shown in Fig. 4(a), the base sample serves as a positive sample and the other four samples can be marked as negative samples. Further, the circulant matrix X can be made diagonal by the Discrete Fourier Transform (DFT) [19]

$$X = C(x_1) = F_1 \text{diag}(F(x_1)) F_1^H \quad (9)$$

where $F(\cdot)$ and $\text{diag}(\cdot)$ denote the DFT and the diagonalization function, respectively. F_1^H denotes the Hermitian transpose of F_1 , which is a constant. Besides, $I = F_1 F_1^H$ and $F_1 u = F(u)$, where u is an arbitrary $n \times n$ matrix.

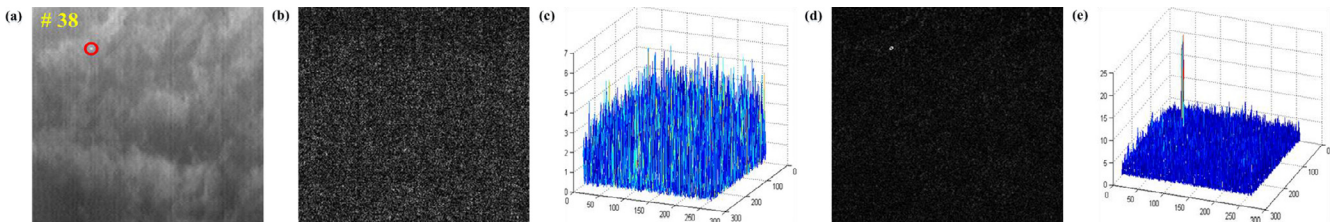


Fig. 1. The effect of image preprocessing over thick cloud. (a) Original image (a target is marked by a red ellipse), (b) Difference between the origin image and the current background, (c) 3D view of (b), (d) Difference between the original image and the updated background, and (e) 3D view of (d).

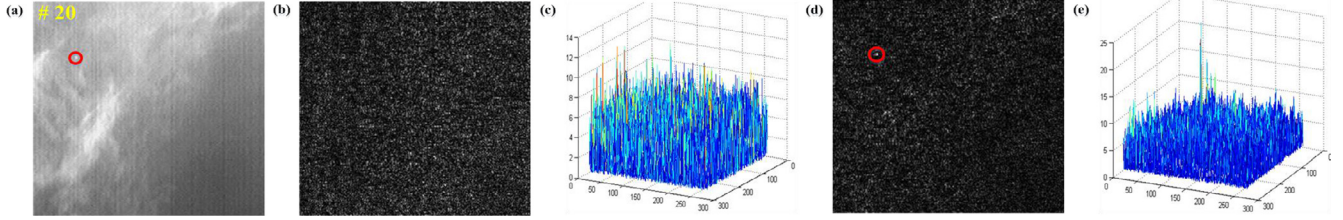


Fig. 2. The effect of image preprocessing over *moving cloud*. (a) Original image (a target is marked by a red ellipse), (b) Difference between the origin image and the current background, (c) 3D view of (b), (d) Difference between the original image and the updated background, and (e) 3D view of (d).

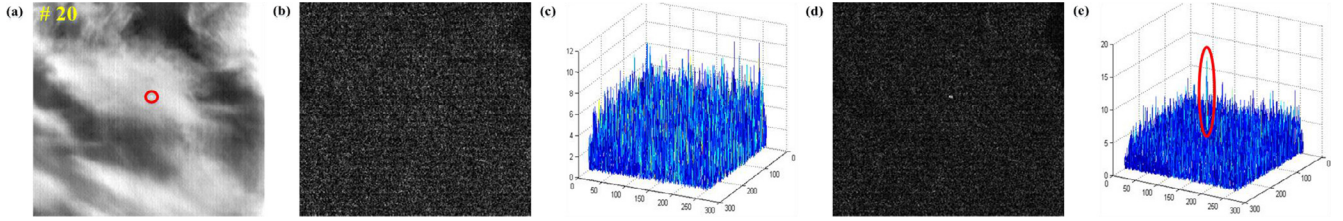


Fig. 3. The effect of image preprocessing over *plane1*. (a) Original image (a target is marked by a red ellipse), (b) Difference between the origin image and the current background, (c) 3D view of (b), (d) Difference between the original image and the updated background, and (e) 3D view of (d). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

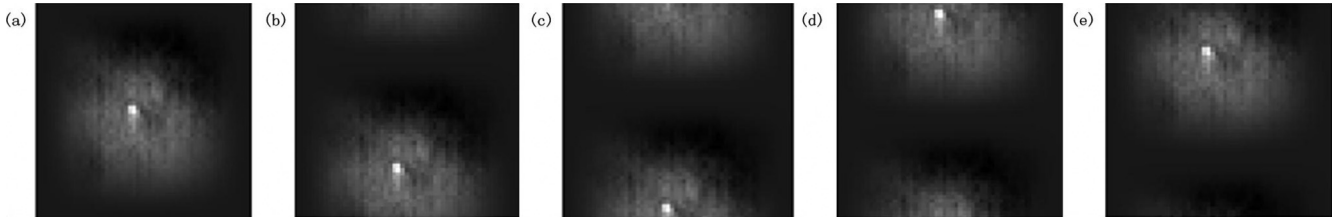


Fig. 4. Several samples obtained by cyclic shift (pixel number $n = 64 \times 64$). (a) Base sample, (b) Negative sample 961, which is obtained by shifting the base sample with 960 pixels, (c) Negative sample 1601, (d) Negative sample 3073, and (e) Negative sample 3713.

3.1.2. Kernelized classifier solution

Moreover, the matrix K in Eq. (6) can be verified to be a circulant matrix, and is made diagonal as

$$K = C(k^{xx}) = F_1 \text{diag}(F(k^{xx})) F_1^H \quad (10)$$

where k^{xx} (with the element $k_i^{xx} = k(x_1, x_i)$ ($i = 1, 2, \dots, n$)) is the first row of the matrix K . Besides, k^{xx} can be expressed as

$$k^{xx} = C(x_1)x_1 = F^{-1}(F(x_1)F(x_1)) \quad (11)$$

where $F^{-1}()$ denotes the inverse DFT.

Therefore, a fast solution [19] to α of Eq. (5) in the Fourier domain is represented by

$$F(\alpha) = F(y)/(F(x_1)F(x_1) + \lambda) \quad (12)$$

3.1.3. Fast detection

When the next frame comes, the testing base sample z_1^{t+1} and the classifier parameter α^{t+1} are updated with a weighted historical information.

$$\alpha^{t+1} = (1 - \rho_1)\alpha^t + \rho_1\alpha_n^t \quad (13)$$

$$z_1^{t+1} = (1 - \rho_2)z_1^t + \rho_2(z_n)_1^t \quad (14)$$

where ρ_1 and ρ_2 are the update factors. $(z_n)_1^t$ is the base sample for new inputs and is set as x_1^t at frame t . α_n^t (calculated by Eq. (12)) denotes the training classifier parameter at frame t . At the first frame, the historical information α^1 and z_1^1 are 0.

Moreover, the responses f^{t+1} for all testing samples can be described as

$$\begin{aligned} f^{t+1} &= \sum_{i=1}^n \sum_{j=1}^n \alpha_j^{t+1} k(x_j^{t+1}, z_i^{t+1}) \\ &= C(k^{xz})^T \alpha^{t+1} \end{aligned} \quad (15)$$

where x_j^{t+1} (its base sample is x_1^{t+1}) denotes the training samples at frame $(t+1)$. k^{xz} (with element $k_j^{xz} = k(z_1^{t+1}, x_j^{t+1})$ ($j = 1, 2, \dots, n$)) is a $n \times 1$ vector. Similarly to Eq. (11), k^{xz} can be expressed as

$$k^{xz} = F^{-1}(F(x_1^{t+1})F(z_1^{t+1})) \quad (16)$$

With Eq. (9), a fast solution to f^{t+1} in Eq. (15) is translate into

$$f^{t+1} = F^{-1}(F(x_1^{t+1})F(z_1^{t+1})F(\alpha^{t+1})) \quad (17)$$

3.2. Deviation analysis

For infrared images, the KCF algorithm realizes target tracking with the image intensity, which is insufficient for dim targets to be distinguished from the cluttered background. According to Eq. (12), the classifier parameter α is only related to x_1 , i.e., the gray intensity or the gray-scale distribution of the base sample. Since the target belongs to the high frequency component of the base sample, its information maps to the high frequency region of α . Likewise, the background tends to map to the low frequency region. However, the information of the edge around the target maps to the high frequency region of classifier parameter α . There-

fore, it is necessary to eliminate the influence of the background edge and get an classifier parameter for dim targets.

Subsequently, it is considerable to filter the background edge (e.g., the edge of cloud or the road boundary) without eliminating the target. Moreover, the key is to choose a suitable filter that can separate the edge and the small target.

3.3. The improved KCF algorithm

To solve the mentioned problem above, a Gaussian Curvature Filter (GCF) [28] algorithm is adopted to handle the base sample, and then is applied to the KCF algorithm to make a tracking result.

3.3.1. The GCF algorithm

The GCF is an adaptive filter that is locally linear variable, and has the characteristic of preserving the edge. Its essence is to calculate the constraint on image surface with the block developable principle, and image denoising can be realized through modifying the pixel value according to the distance between this pixel and the tangent plane of its neighbor pixel. Given a base sample region p in the current frame, and its size is set as 7 times (just for explanation) of the target template, containing 8×8 pixels. For one pixel (i, j) ($i, j = 1, 2, \dots, 56$) in p , the first filtering result q_1 is formulated as

$$q_1(i, j) = p(i, j) + \min\{|d_l|, l = 1, \dots, 8\} \quad (18)$$

where d_l is defined as

$$d_1 = (p(i-1, j) + p(i+1, j))/2 - p(i, j) \quad (19)$$

$$d_2 = (p(i, j-1) + p(i, j+1))/2 - p(i, j) \quad (20)$$

$$d_3 = (p(i-1, j-1) + p(i+1, j+1))/2 - p(i, j) \quad (21)$$

$$d_4 = (p(i-1, j+1) + p(i+1, j-1))/2 - p(i, j) \quad (22)$$

$$d_5 = p(i-1, j) + p(i, j-1) - p(i-1, j-1) - p(i, j) \quad (23)$$

$$d_6 = p(i-1, j) + p(i, j+1) - p(i-1, j+1) - p(i, j) \quad (24)$$

$$d_7 = p(i, j-1) + p(i+1, j) - p(i+1, j-1) - p(i, j) \quad (25)$$

$$d_8 = p(i, j+1) + p(i+1, j) - p(i+1, j+1) - p(i, j) \quad (26)$$

Then, q_1 is regarded as an input of the next filtering with Eq. (18). Besides, the iteration number is denoted by m and the final resulting q_m of the GCF algorithm can be described as

$$q_m = G_m(p) \quad (27)$$

where $G_m(\cdot)$ denotes m times iterative computations of Eqs. (19)–(26).

Therefore, the classifier parameter α in the Fourier domain is expressed as

$$F(\alpha) = F(y)/((F(g_1))^2 + \lambda) \quad (28)$$

where g_1 denotes the difference between the base sample x_1 and its filtering result.

$$g_1 = x_1 - G_m(x_1) \quad (29)$$

Further, The effect of the GCF algorithm is shown in Fig. 5.

Occasionally, infrared small targets submerge in the cluttered background of the original sample, even if it is processed by the SVD-based algorithm (see Fig. 2(d)–(e)). As shown in Fig. 5(c), its corresponding classifier parameter α in the Fourier domain contains little high frequency information, meaning that the object information is weak. With the GCF algorithm, the edge around the target in the base sample is restrained (see Fig. 5(b)), and the

target's information maps to the high frequency region (center region) of classifier parameter α in the Fourier domain, which is indicated in Fig. 5(d). To sum up, it is possible to obtain a classifier parameter α for a small target, owing to the fact that the GCF algorithm can preserve the edge well. Further, a response map of the dim target can be calculated via Eq. (17).

3.3.2. The SVD-IKCF algorithm

To deal with the excursion problem, the GCF algorithm is adopted to handle the base sample for the KCF algorithm. Besides, the difference between the original image and the filtering result is regarded as an input (i.e., a training sample set) of the KCF tracker. The SVD-IKCF algorithm is described as follows.

Step 1: Tracker initialization

- (1) At the first frame, target position is initialized through a manual selection. The size of the target template is set as 8×8 , and the center denotes the target position.
- (2) Tracking parameter. The size of the sample region, the weight factor γ in Eq. (4), the regularization λ in Eq. (5), the prior y in Eq. (7) and the iteration number m in the GCF algorithm are set at the beginning.

Step 2: Target enhancement. Infrared backgrounds are suppressed and targets are enhanced by the SVD-based algorithm.

Step 3: At frame t , the base sample x_1^t is processed via the GCF algorithm and the training base sample g_1^t is expressed as

$$g_1^t = x_1^t - G_m(x_1^t) \quad (30)$$

Therefore, a classifier parameter α_n^t for the dim target is calculated via Eq. (28),

$$F(\alpha_n^t) = F(y)/((F(g_1^t))^2 + \lambda) \quad (31)$$

Step 4: While the next frame comes, the response map f^{t+1} for all inputs z^{t+1} (its base sample is z_1^{t+1} and the other testing sample are obtained by shifting z_1^{t+1}) is formulated as

$$f^{t+1} = F^{-1}(F(g_1^{t+1})F(z_1^{t+1})F(\alpha^{t+1})) \quad (32)$$

where g_1^{t+1} is the training base sample at frame $(t+1)$. α^{t+1} and z_1^{t+1} have been updated with the historical information, i.e., Eqs. (13), (14).

Step 5: Target position is obtained via the target response map f^{t+1} .

Step 6: Continuing to deal with the next frame by Step 2.

The flow chart of the SVD-IKCF algorithm is shown in Fig. 6.

4. Experimental results and analysis

The SVD-IKCF algorithm performance is evaluated over six infrared sequences, containing kinds of complex backgrounds. Seq.1–6 are named *tank*, *plane1*, *plane2*, *moving cloud*, *thick cloud* and *plane3*, respectively. As shown in Figs. 8–10, Seq.2–4 are obtained by capturing the central 256×256 pixels from images taken by a 640×480 pixels uncooled Infrared Focal Plane Array (IRFPA) camera.

In Seq.1, the target motion is unstable, which is caused by camera shake. Its size is from 3×3 pixels to 7×7 pixels.

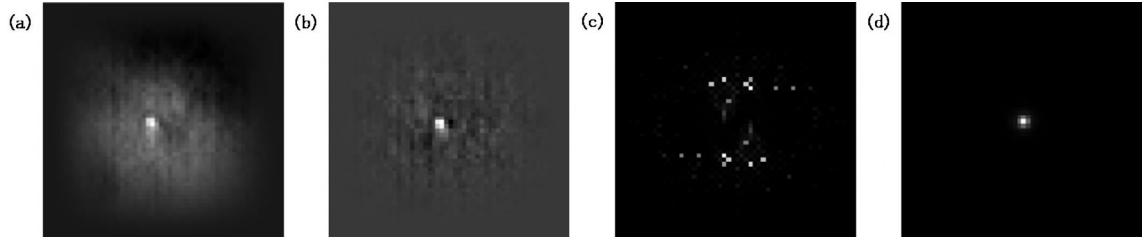


Fig. 5. The effect of the GCF algorithm. (a) Base sample, (b) base sample with GCF, (c) original $F(\alpha)$, and (d) $F(\alpha)$ with GCF.

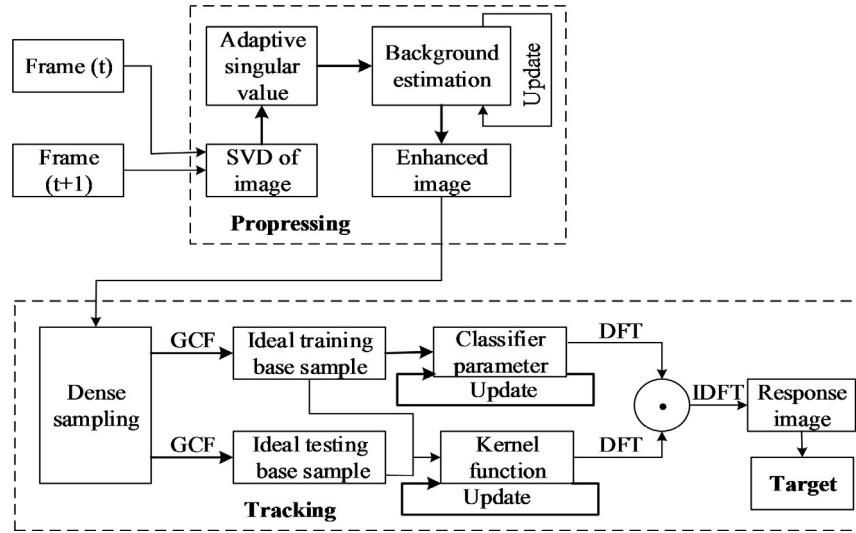


Fig. 6. Flow chart of dim moving target tracking algorithm.

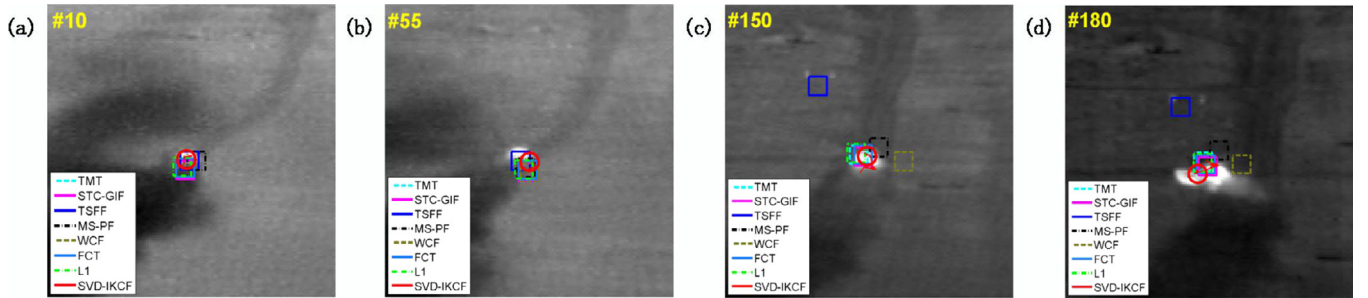


Fig. 7. Qualitative results of the 5 trackers over *tank*. Best viewed on color display.

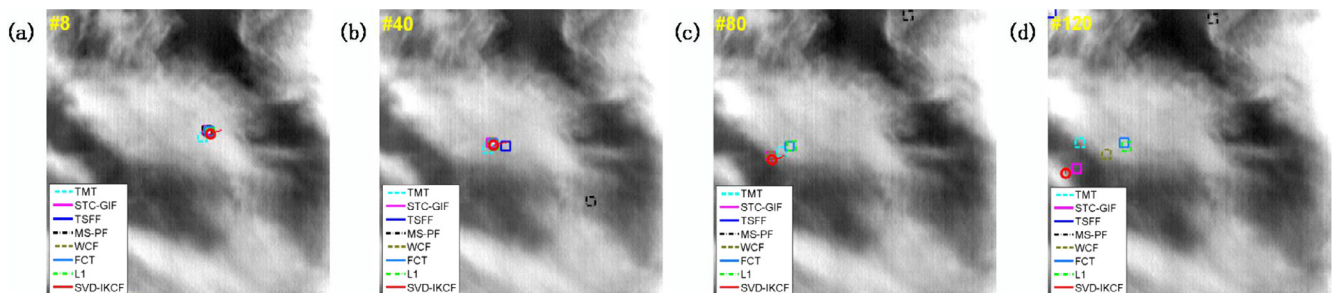


Fig. 8. Qualitative results of the 5 trackers over *plane1*, in which the edge is strong. Best viewed on color display.

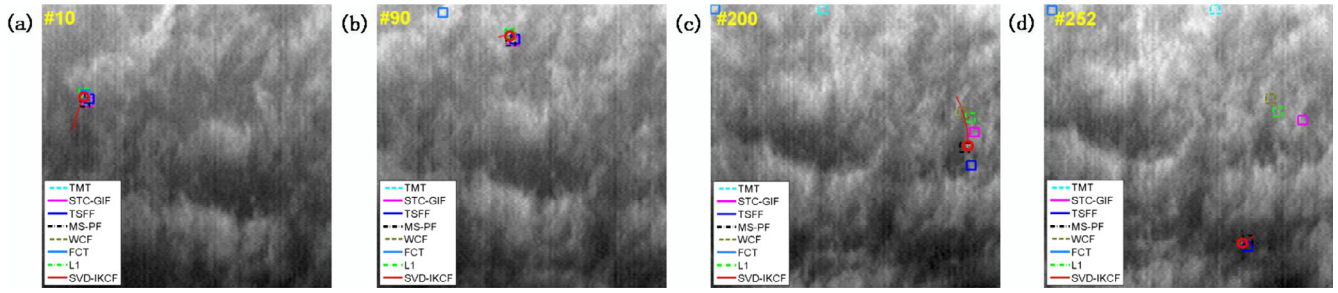


Fig. 9. Qualitative results of the 5 trackers over *thick cloud*. Best viewed on color display.

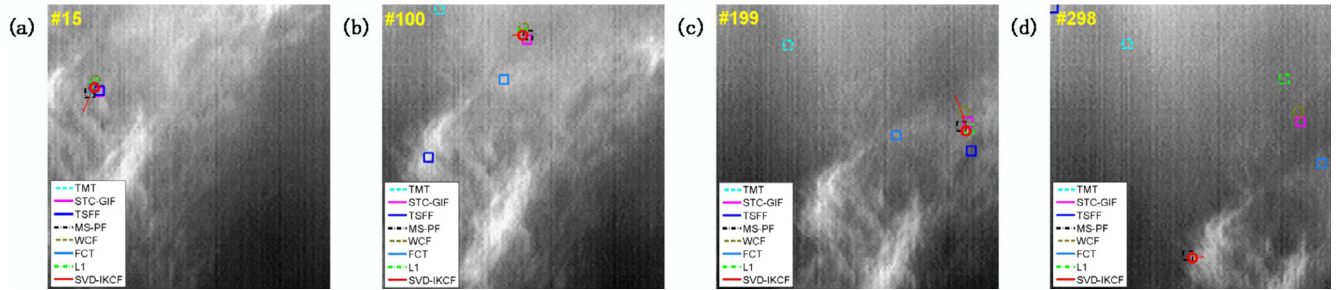


Fig. 10. Qualitative results of the 5 trackers over *moving cloud*. Best viewed on color display.

In Seq.2, the edge of cloud is so strong that the dim target's SNR is lower than 1.5. Besides, the target moves across the background edge in several dozens of images. The sequence length is 120.

In Seq.3, a small target is added manually, and its size is set as 3×3 pixels. Besides, the target undergoes cluttered background, and moves in a curve with a large instant speed (i.e., 8 pixels per frame). The sequence contains 300 frames.

In Seq.4, the property of the dim target is the same as that in Seq.3. The only difference is that Seq.4 contains moving cloud, which moves fast than that in Seq.2.

Seq.5 is obtained by capturing the central 400×450 pixels from images taken by a medium wave infrared camera. The sequence contains a target with the speed of about 8 pixels per frame, and its length is 50.

In Seq.6, a dim target under the space background appears in the upper right corner at the beginning and follows a downward diagonal path. Besides, some false objects appears in the image sequence. The sequence includes 500 frames.

Experiments are implemented with MATLAB 2013b on a PC with a 3.3GHZ Intel-Core CPU and 4 GB memory. The SVD-IKCF algorithm is tested and compared with seven algorithms, including the improved TMT algorithm [3], the TSFF algorithm [7], the MS-PF algorithm [9], the FCT algorithm [14], the ℓ_1 -based algorithm [15], the WCF algorithm [17] and the STC-GIF algorithm [18].

4.1. Experimental setup

The size of the tracking box is 8×8 pixels (its center denotes the target position), and the size of the sample region is set as 7 times of the tracking box, in order to adapt to the speed of the small target. However, the value can be 3 if targets move slowly. In the SVD algorithm, c is set as 90, which is evaluated by experiments to enhance targets well. The weight factor γ in Eq. (4) is 0.7, which given a high weight to the current background. In Eq. (5), λ is 0.01. In Eq. (7), $\sigma_1 = 2.25$ and $\beta = 1$, in order to prevent over-fitting in searching for the object. In Eqs. (13) and (14), ρ_1 and ρ_2 are set as the same value 0.075, which given a high weight to the historical information. In the GCF algorithm, the iteration number m in

Eq. (27) is set as 10, which gives consideration to the algorithm speed and the convergence.

4.2. Qualitative comparison

Qualitative results of eight tracking algorithms are shown in Figs. 7–12. In the *tank* sequence, the camera undergoes a sever shake. The TSFF algorithm has frequent detection error (e.g., #150 and #180), due to continuous motion of background pixels. The WCF drifts to the background at frame #150 and #180. The FCT algorithm and the ℓ_1 algorithm can track the target well, owing to the fact that the target feature is more discriminative and the ground background contains less clutter for Seq.1. Benefiting from the dense sampling, the STC-GIF and the presented algorithm can obtain large amount of target information, and the tracking result is satisfactory. In the *plane1* sequence, the target submerges in bright cloud with strong radiation. As shown in Fig. 8, only the SVD-IKCF algorithm can track the target well over all images. The classic TSFF algorithm loses the target frequently. The TMT algorithm and the MS-PF algorithm fail to tracking the target when it moves cross the strong edge, and the problem is unrecoverable at frame #80. In the *thick cloud* sequence, the target submerges in the cluttered background and its instant speed is up to 8 pixels per frame at frame #200. The TSFF algorithm, the MS-PF algorithm and the presented algorithm can detect the target at frame #252. Besides, The TMT algorithm drifts to the background at frame #200. The FCT algorithm loses the target at frame #90 and the accumulative error becomes larger and larger, which is a bottleneck of most tracking algorithms without using detection. The ℓ_1 algorithm and the STC-GIF algorithm is far away from the target when the target move fast (e.g., frame #200) and the tracking excursion can't be eliminated. In the *moving cloud* sequence, the property of the dim target is the same as that in the *thick cloud* sequence. The TSFF algorithm tends to detect the moving edge as a target (e.g., frame #100). Both the TMT algorithm and the FCT algorithm lose the target after frame #100. The ℓ_1 algorithm and the STC-GIF algorithm drift to the background at frame #199, for the reason that the dim target moves so fast that it is out of the

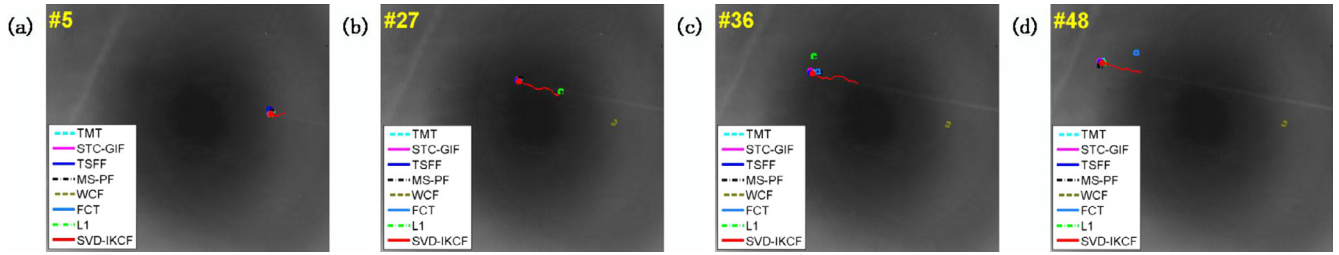


Fig. 11. Qualitative results of the 5 trackers over *plane2*, with large image size. Best viewed on color display.

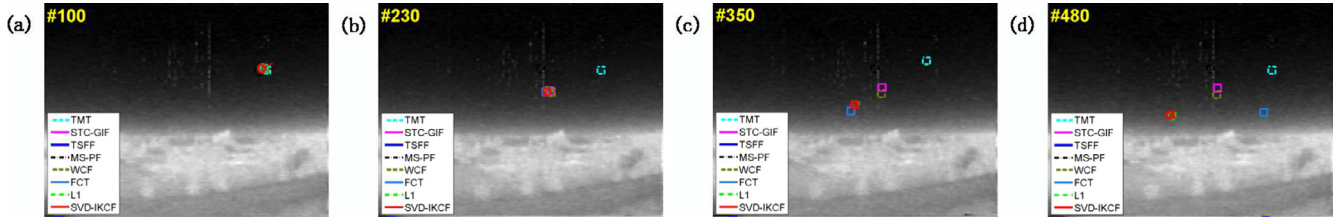


Fig. 12. Qualitative results of the 5 trackers over *plane3*. Best viewed on color display.

search scope. Increasing the search scope will bring in much background clutter to the sample set, which makes tracking migrate. Both the MS-PF algorithm and the SVD-IKCF algorithm perform well in the entire sequence. In the *plane2* sequence, the image size is large. The ℓ_1 algorithm and the WCF algorithm fail to estimate an accurate position after frame #27 and the FCT loses the target at frame #48. In the *plane3* sequence, the background has low gray value and the target gradually disappears behind the horizon. The tracking result of TSFF algorithm are unacceptable. The TMT algorithm drifts to the background after frame #230. Several algorithms (e.g., the MS-PF algorithm and the STC-GIF algorithm and the WCF algorithm) detect a false target at frame #350. Yet, the ℓ_1 algorithm and the SVD-IKCF algorithm obtain satisfactory results.

The SVD-IKCF algorithm has done well in tracking the dim moving target regardless of the ground background, the cloud background or the space background, for the reason that the dense sampling is rationally utilized. Besides, the kernel trick deals with the nonlinear noise effectively and the GCF algorithm does well in weakening the effect of the background edge around the dim target.

4.3. Quantitative comparison

For further quantitative analysis, several similar standard evaluations in [16] are adopted. They are Center position Error (CLE), Success Rate (SR) and Frame Per Second (FPS). The CLE denotes the Euclidean distance d between an estimated target and a real target. The SR is defined as n_1/s , where s is the frame number of a sequence and n_1 is the frame number on condition that $d < 6$.

Therefore, the CLE and the SR represent the accuracy while the FPS represents the algorithm speed. Then, a group of experimental results is shown in Table 1 and Fig. 13.

In Fig. 13, the vertical axis represents the logarithm of the CLE value, for a suitable scale. $\log$ represents the logarithm operator and the line ' $y = \log(6)$ ' is a boundary (equal to that the CLE value is 6). As shown in Fig. 13(a), it is obvious that most algorithms can detect the target well. Nevertheless, the CLE curves of TSFF algorithm and the WCF algorithm have a large fluctuation. In Fig. 13(b), only the STC-GIF and the SVD-IKCF algorithms obtain satisfactory results and the other algorithms lose the target at most frames, for the reason that the *plane1* sequence contains targets with a low SNR and much background edge. With large amount of background clutter, the target template in the TMT algorithm can't be updated ideally, leading to the cumulative error (see Fig. 13(b)–(d)). It is indicated in Fig. 13(e) that the WCF algorithm always drifts to the background. Generally, the STC-GIF algorithm is adaptive to kinds of backgrounds. However, it is difficult to adapt to the situation that the target moves fast while the current search region is not large enough, which is shown in Fig. 13(c) and (d). Besides, the tracking excursion is unrecoverable after frame #200, and the TMT algorithm and the FCT algorithm fail to detect the target with large CLE values. As shown in Fig. 13(f), most algorithms lose the target after frame #250. Both the ℓ_1 algorithm and the SVD-IKCF algorithm have small CLE values.

The CLE result of the SVD-IKCF algorithm over all infrared sequences is satisfactory, demonstrating that the presented algorithm can track the dim moving target well.

Table 1

Success Rate (SR) (%) and average Frame Per Second (FPS). **Bold** fonts indicate the best performance while the *italic* fonts indicate the second best ones. The total number of evaluated frames is 1470.

Sequence	TMT	STC-GIF	TSFF	MS-PF	WCF	FCT	ℓ_1	SVD-IKCF
<i>tank</i>	0.98	0.99	0.58	0.91	0.5	0.98	0.94	0.99
<i>plane1</i>	0.15	0.77	0.21	0.24	0.69	0.54	0.58	1
<i>thick cloud</i>	0.29	0.67	0.97	0.61	0.52	0.1	0.62	1
<i>moving cloud</i>	0.17	0.66	0.96	0.58	0.69	0.1	0.65	1
<i>plane2</i>	0.18	1	0.34	0.54	0.1	0.18	0.34	1
<i>plane3</i>	0.16	0.5	0	0.14	0.51	0.62	0.95	1
Average SR	0.32	0.77	0.51	0.5	0.5	0.42	0.68	0.99
Average FPS	27	55	8.5	21	34	40	13.5	75

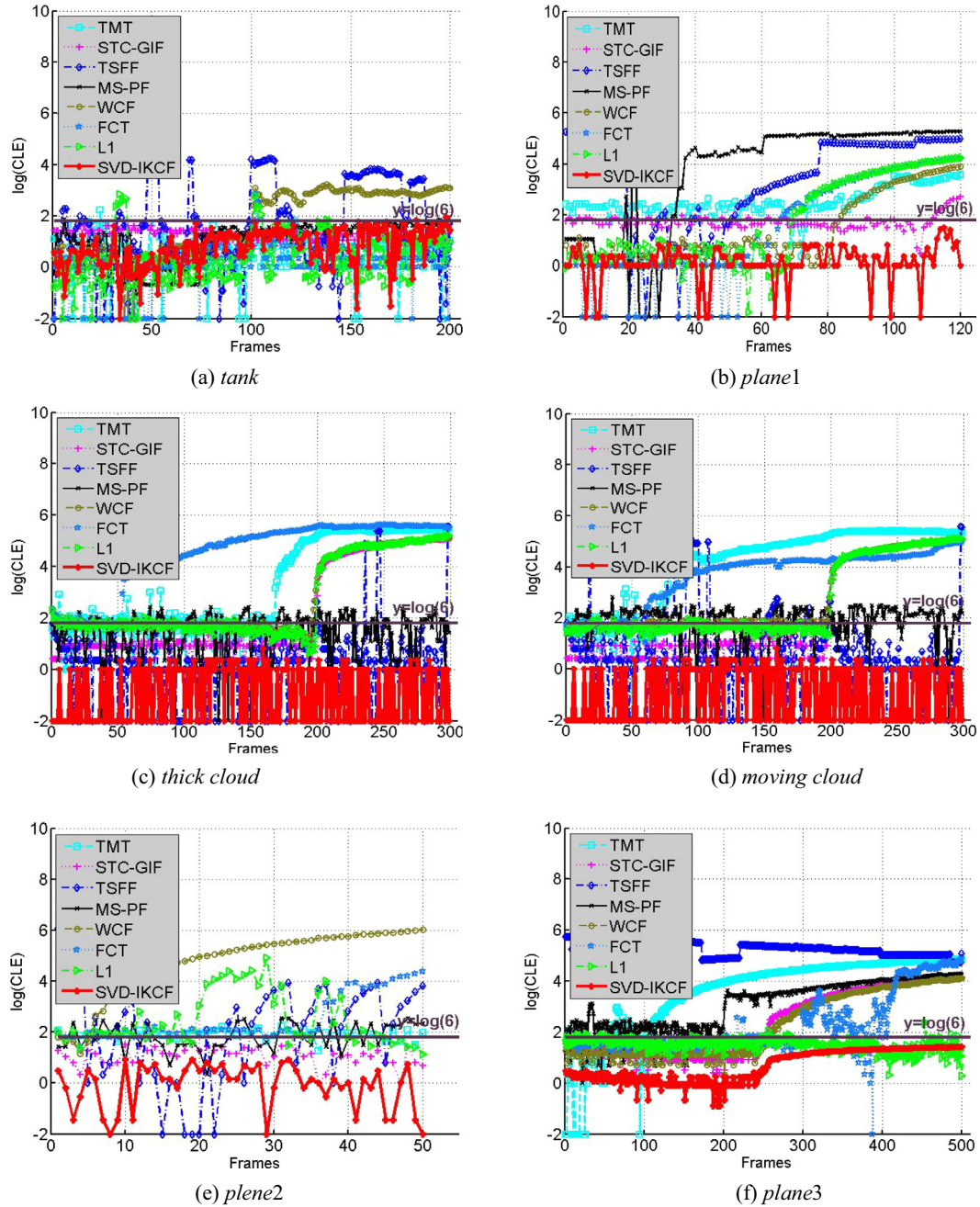


Fig. 13. Logarithm of center position error.

As shown in Table 1, the success rate of the TMT algorithm over 5 sequences (except *tank*) is low (at most 29%), due to the fact that this algorithm is sensitive to the environment. The TSFF algorithm has a modest success rate over Seq.2 and Seq.3 (i.e., 97% and 96%). Nevertheless, it is time-consuming (the FPS is 8.5) and is greatly influenced by the background clutter, e.g., the FPS over *plane3* is 0. The MS-PF algorithm can't obtain a stable gray statistics when the target submerges into the bright cloud (e.g., Seq.2). The FCT algorithm has a small success rate when the target submerges in the cluttered background (Seq.2 or Seq.3), resulting in an inaccurate classification. Besides, the ℓ_1 algorithm utilizes the appearance model in the FCT algorithm with the ℓ_1 constraint, and its average SR value is greater than that of the FCT algorithm. Since the STC-GIF algorithm searches the target in a local region, a fast target's position may not be estimated accurately. For example, in the *thick*

cloud sequence and the *moving cloud* sequence, the success rate are 67% and 66%, which are smaller than that in several other sequences. For fair comparison, the sample size in the STC-GIF algorithm and the SVD-IKCF algorithm are set as a same value over all infrared sequence. In Seq.6, the sample size in both algorithms reduces to three times of the tracking box, helping to improving the presented algorithm's speed without losing the tracking accuracy.

In contrast, the SVD-IKCF algorithm (the average SR is rounded to 1) does well in tracking a dim moving target. On the one hand, target is enhanced by the SVD algorithm, which is an efficient method to reconstruct the background. On the other hand, the GCF algorithm weakens the negative effect of the background edge for the KCF algorithm. Experimental results insist that the SVD-IKCF algorithm can run at a high FPS value (the average FPS is

75), due to the fact that this presented algorithm deals with small targets in the Fourier domain. In addition, the GCF algorithm contains only a few multiply-add operations, which can handle the base sample in a short time. Moreover, the algorithm speed can be greatly improved by decreasing the searching scope, provided that the target moves slowly.

In terms of subjective vision and objective evaluation, the SVD-IKCF algorithm is superior to several other algorithms for infrared dim and small target tracking.

5. Conclusions

The performance of the infrared detection system is often influenced by cluttered backgrounds. Therefore, an effective and high-speed algorithm which takes advantage of the SVD-based algorithm and the improved KCF algorithm was presented for infrared dim-small target tracking. The enhanced image sequence, serving as an input of the improved KCF algorithm, was obtained by the SVD algorithm. In the nonlinear KCF algorithm, a classifier parameter of small targets can be calculated by processing the base sample with the GCF algorithm, which can weaken the negative effect of the background edge. At last, a response map was obtained and the target position was determined by finding a maximal response value. Experimental results demonstrate that the presented algorithm performs quite well in achieving both high accuracy and speed, and, broadly, makes great contributions on laying the foundation for future recognition.

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